



Probability and Bayes' Theorem

From basic chance to updated risk estimates

Simple explanations • worked examples • practical use cases



1. Basic probability

Probability measures how likely an event is. It ranges from 0 to 1, or from 0% to 100%.

$$P(\text{Event}) = \text{favourable outcomes} / \text{total possible outcomes}$$

If 20 of 1,000 transactions are fraudulent, the estimated probability of fraud is $20/1,000 = 0.02$, or 2%.

- 0 means impossible under the model.
- 1 means certain under the model.
- Probabilities of all mutually exclusive outcomes together total 1.

2. Conditional probability

Conditional probability is the chance of event A after we know that event B occurred.

$$P(A | B) = P(A \text{ and } B) / P(B)$$

Suppose 100 transactions are international and 12 are fraudulent. Among international transactions, the fraud probability is $12/100 = 12\%$. This may differ from the overall fraud rate.

The vertical bar in $P(\text{Fraud} | \text{International})$ means 'Fraud given that the transaction is international.'

3. Bayes' theorem

Bayes' theorem reverses a condition. It combines a prior probability with new evidence to produce an updated probability.

$$P(A | B) = [P(B | A) \times P(A)] / P(B)$$

Fraud-alert example

Assume fraud prevalence is 1%, an alert detects 90% of fraud, and falsely alerts on 5% of genuine transactions.

$$P(\text{Fraud} | \text{Alert}) = (0.90 \times 0.01) / [(0.90 \times 0.01) + (0.05 \times 0.99)] \approx 15.4\%$$

Even with a strong detector, most alerts may be genuine transactions because fraud is rare. This is the base-rate effect.

4. Banking use cases and cautions

Use case	Evidence	Updated decision
Fraud screening	Location, device, amount	Recalculate fraud risk
Credit risk	Income, repayment history	Update probability of default
Customer churn	Complaints, inactivity	Prioritize retention action
AML review	Transaction pattern, counterparties	Prioritize cases for investigation

What probability does not mean

- A 20% risk does not mean the event will occur exactly once in every five individual cases.
- A model probability is not automatically correct; it depends on data quality and assumptions.
- Conditional probability does not establish causation.
- Bayesian updates can be misleading when the prior or evidence rates are inaccurate.